Soru $x(t) = 2\,\mathrm{sinc}(40t)$ işareti transfer fonksiyonu $H(f)$ olan bir kanaldan iletilmektedir.

$y(t) = 20\,\mathrm{sinc}(40t - 200)$ old. göre,

a) $H(f)$'yi bulun Genlik ve faz spektrumlarını çizin

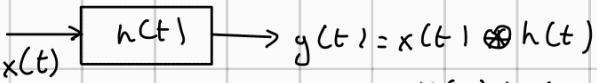
b) $h(t)$'yi bulun

c) Bu sistem bozulmasız mıdır?

$x(t) = t\,\mathrm{sinc}^2(t)$ için $y(t) = ?$

ol) $x(t) = 2\,\mathrm{sinc}(40t)$ için, çıkışın enerji spektral yoğunluğunu bulun

e) Çıkış işaretinin enerjisini bulun.



$\mathcal{E}_x(f) = |X(f)|^2$

$Y(f) = X(f)\,H(f)$

$\mathcal{E}_Y(f) = |H(f)|^2 \mathcal{E}_x(f)$

$x(t - t_0) \rightarrow X(f)\, e^{-j2\pi f t_0}$

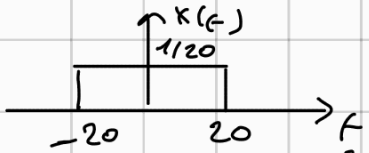
$A z \sin(tz) \xleftrightarrow{F} A\,\Pi(f/z)$

a) $x(t) = 2\,\mathrm{sinc}(40t) \xleftrightarrow{F} X(f) = \frac{1}{20}\,\Pi(f/40)$

$z = 40$

$A \cdot z = 2$

$A = 1/20$



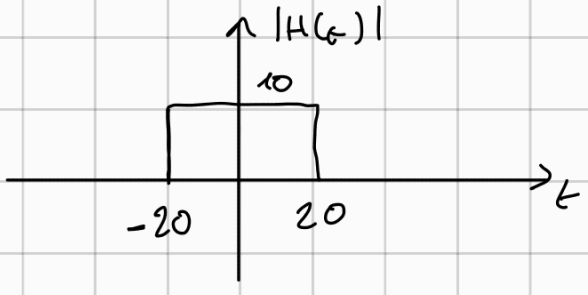
$y(t) = 20\,\mathrm{sinc}(40(t-5)) \xrightarrow{F} Y(f) = \frac{1}{2}\,\Pi(f/40)\, e^{-j2\pi f 5}$

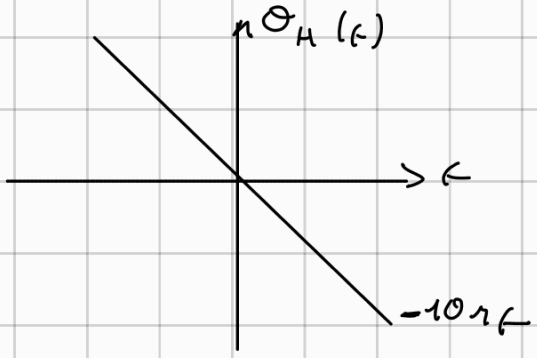
$z = 40$

$A = 1/2$

$H(f) = \frac{Y(f)}{X(f)} = 10\,\Pi(f/40)\, e^{-j2\pi f 5}$

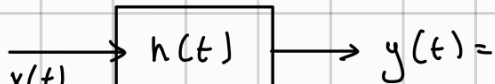
yine karedalar bu geçer





b) $h(t) = 400\,\mathrm{sinc}(40(t-5))$

c) Bozulmasızdır. (Çünkü giriş işareti için) genlik spektrumu sabit, faz spektrumu lineerdir



$|H(f)| = k \qquad \Theta_H(f) = -2\pi f t_d \qquad k = 10 \qquad t_d = 5$

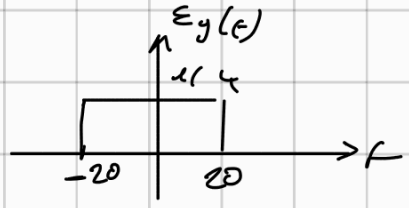
$x(t) = t\,\mathrm{sinc}^2(t)$

$y(t) = 10x(t-5) = 10\,(t-5)\,\mathrm{sinc}^2(t-5)$

d) $x(2) = 2\,\mathrm{sinc}(40t)$

$\mathcal{E}_x(f) = |X(f)|^2 = \frac{1}{400}\,\Pi(f/40)$

$= \frac{1}{4}\,\Pi(f/40) \longrightarrow$



e) $E_y = \int_{-\infty}^{\infty} \mathcal{E}_y(f)\,df$

$= \int_{-20}^{20} 1/4\,df = \frac{1}{4}\cdot 40 = 10\,j$

Soru: Bir LTI sistem olarak modellenen kanalın dürtü cevabı

$h(t) = 5\,\delta(t-10)$ olarak verilmektedir

a) Bu sistemin Transfer fonksiyonunu bulun

Genlik ve faz spektrumlarını çizin.

Bu sistem bozulmasız mıdır? Neden??

b) $x(t) = 10 + 10t\,\mathrm{sinc}(t)$ işareti uygulanırsa çıkışı nedir

c) $x(t) = 20\cos(2000\pi t) + 10\sin(4000\pi t)$ ise $x(t)$'nin güç yoğunluk fonksiyonunu (power spectral density) bulun

d) c şıkkındaki $x(t)$ için $y(t)$'yi ve $y(t)$'nin GYF'sini bulun

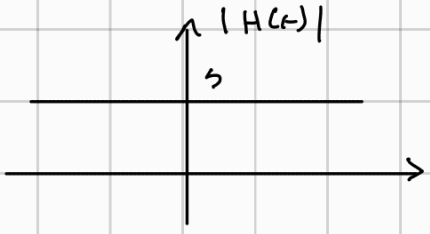
e) $y(t)$'nin özilişki fonksiyonunu bulun

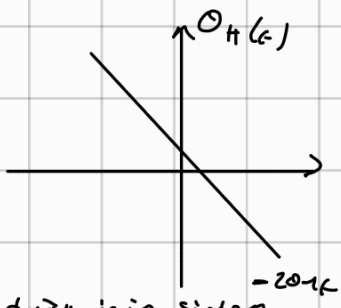
f) $y(t)$'nin gücünü hem özilişki fonkdan hemde GYF'den bulun.

a) $h(t) = 5\delta(t-10) \xrightarrow{F} H(f) = 5e^{-j2\pi 10f}$

$H(f) = |H(f)|\,e^{j\theta_H(f)}$

$|H(f)| = 5 \qquad \theta_H(f) = -2\pi 10f$





Genlik spektrumu sabit, faz spektrumu lineer olduğu için sistem bozulmasızdır

b) $x(t) = 10 + 10\,t\,\mathrm{sinc}(t)$

$y(t) = k\,x(t-t_0) = S_x(t-10)$

$= 50 + 50(t-10)\sin(t-10)$

$x(t) = \sum_n x_n e^{j2\pi n f_0 t}$

$R_x(z) = \sum_n |x_n|^2 e^{j2\pi n f_0 t}$ $\quad \cos(\theta) = \frac{1}{2}\left[e^{-j\theta} + e^{j\theta}\right]$

$S_y(f) = \sum_n |x_n|^2 \delta(f - n f_0)$

c) $x(t) = 20\cos(2\pi 1000 t) + 10\sin(2\pi 2000 t)$

$= \underbrace{10}_{x_1} e^{-j2\pi 1000 t} + \underbrace{10}_{x_1} e^{j2\pi 1000 t} + \underbrace{\frac{10}{2j}}_{x_2} e^{j2\pi 2000 t} \underbrace{- \frac{10}{2j}}_{x_{-2}} e^{-j2\pi 2000 t}$

$S_x(f) = 100\,\delta(f-1000) + 100\,\delta(f+1000) + 25\,\delta(f-2000) + 25\,\delta(f+2000)$

d) $y(t) = S_x(t-10) = 100\cos(2\pi 1000(t-10)) + 50\sin(2\pi 2000(t-10))$

$S_y(f) = |H(f)|^2 S_x(f) = 25\,S_x(f) = 2500\left[\delta(f-1000) + \delta(f+1000)\right] + 625\left[\delta(f-2000) + \delta(f+2000)\right]$

e) $R_y(z) = 5000\cos(2\pi 1000 t) + 1250\cos(2\pi 2000 t)$

f) $P_y = \int_{-\infty}^{\infty} S_y(f)\,df = R_y(0)$

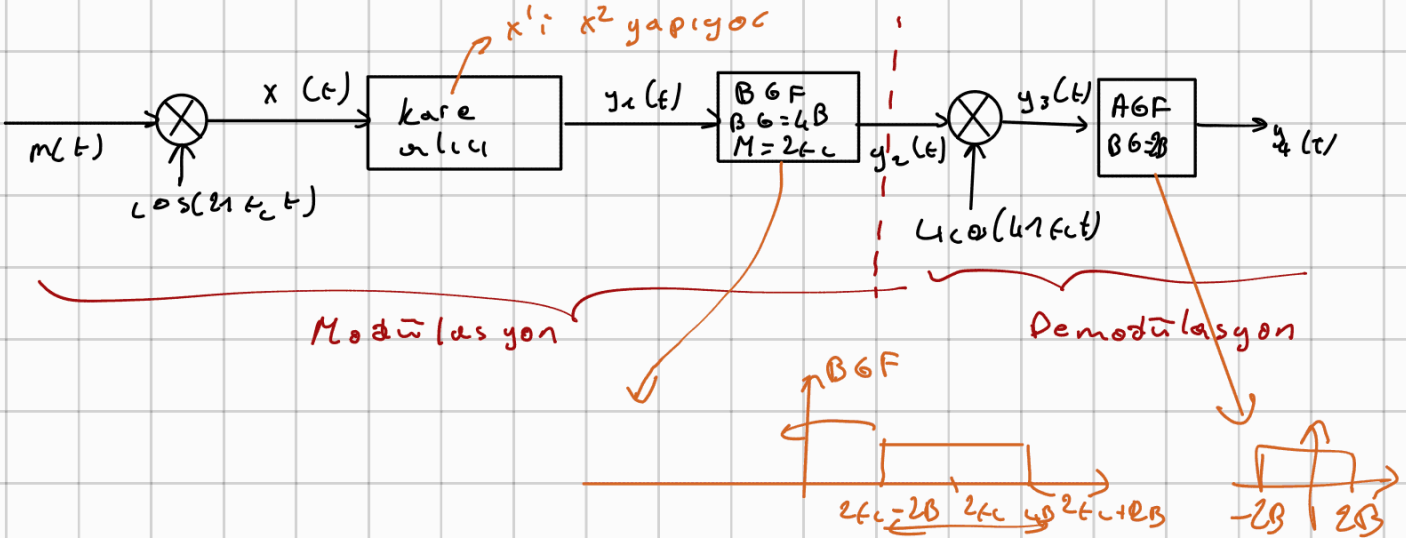
$R_y(z) = \int_{-\infty}^{\infty} S_y(f) e^{j2\pi f z}\,df$

$P_y = R_z(0) = 6250$

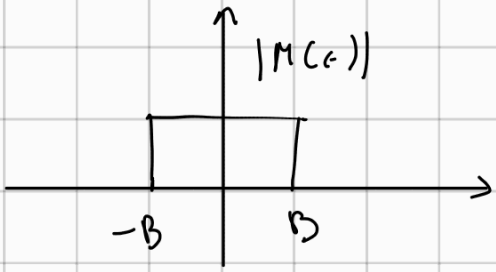
Soru Fourier spektrumu $M(f) = \Pi(f/2B)$ olan mesaj işareti aşağıda verilen sistemden geçmektedir.

$x(t), g_1(t), g_2(t), y_3(t)$ ve $g_4(t)$'yi zaman ve frekans domainde ifade ediniz spektrumlarını çiziniz

$f_c >> B$



$$M(f) = \Pi\left(\frac{f}{2B}\right)$$



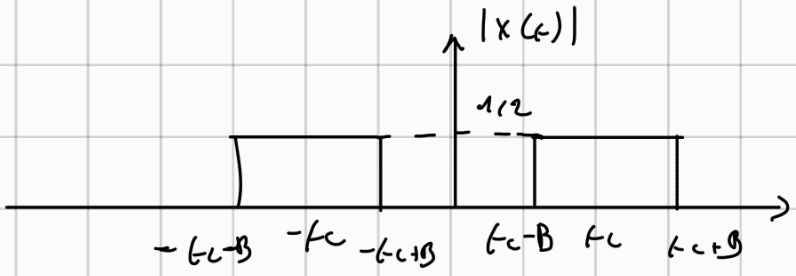
$$M(f) = \Pi\left(\frac{f}{2B}\right)$$

$$\downarrow m(t) = 2B\,\mathrm{sinc}(2Bt)$$

$$= 2B\,\mathrm{sinc}(2Bt)\cos(2\pi f_c t)$$

$$X(f) = \frac{1}{2}M(f-f_c) + \frac{1}{2}M(f+f_c)$$

$$X(f) = \frac{1}{2}\Pi\left(\frac{f-f_c}{2B}\right) + \frac{1}{2}\Pi\left(\frac{f+f_c}{2B}\right)$$



$$y_1(t) = x^2(t) = [m(t)\cos(2\pi f_c t)]$$

$$= 4B^2\,\mathrm{sinc}^2(2Bt)\cos^2(2\pi f_c t)$$

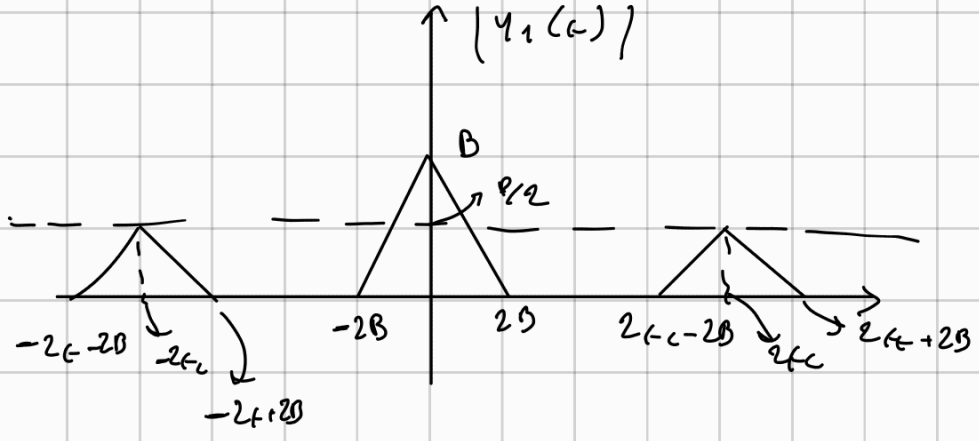
$$\cos^2(2\pi f_c t) = 1 + \frac{1}{2}\cos(2\pi 2f_c t)$$

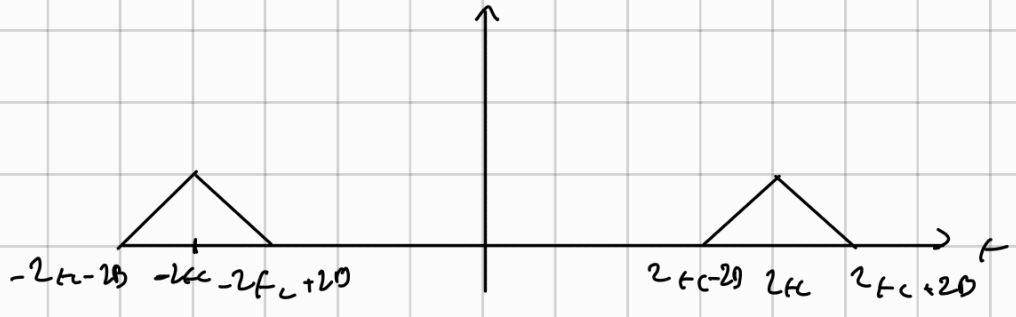
$$= 2B^2\,\mathrm{sinc}^2(2Bt) + 2B^2\,\mathrm{sinc}^2(2Bt)\cos(2\pi 2f_c t)$$

$$z = 2B$$

$$A = B$$

$$y_1(f) = B\,\Lambda\left(\frac{f}{2B}\right) + \frac{B}{2}\Lambda\left(\frac{f-2f_c}{B}\right) + \frac{B}{2}\Lambda\left(\frac{f+2f_c}{2B}\right)$$





$$y_2(t) = \frac{B}{2}\Lambda\left(\frac{f-2f_c}{2B}\right) + \frac{B}{2}\Lambda\left(\frac{f+2f_c}{2B}\right)$$

$$y_2(t) = 2B^2\,\mathrm{sinc}^2(2Bt)\cos(2\pi 2f_c t)$$ ← modüle edilmiş işaret

$$y_3(t) = 4y_2(t)\cos(2\pi 2f_c t)$$

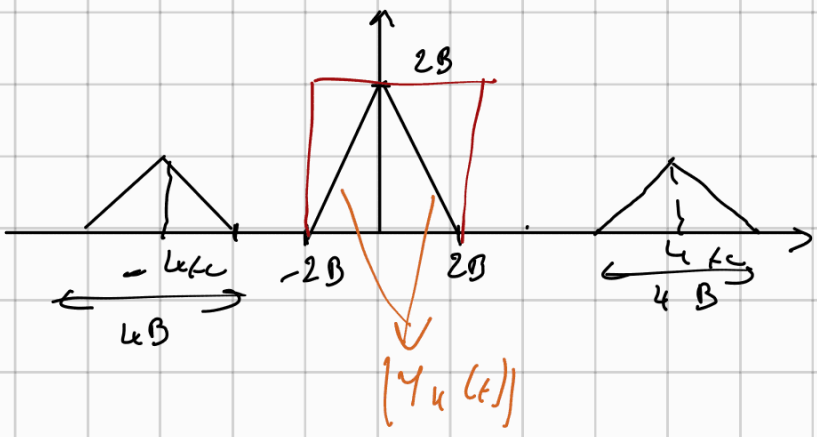
$$= 8B^2\,\mathrm{sinc}^2(2Bt)\cos^2(2\pi 2f_c t)$$

$$\cos^2 = \frac{1}{2} + \frac{1}{2}\cos(2\pi 4f_c t)$$

$$= 4B^2\,\mathrm{sinc}^2(2Bt) + 4B^2\,\mathrm{sinc}^2(2Bt)\cos(2\pi 4f_c t)$$

$z = 2B,\ A = 2B$

$$y_3(t) = 2B\Lambda\left(\frac{f}{2B}\right) + B\Lambda\left(\frac{f-4f_c}{2B}\right) + B\Lambda\left(\frac{f+4f_c}{2B}\right)$$



$$Y_4(f) = 2B\Lambda\left(\frac{f}{2B}\right)$$

$$y_4(t) = 4B^2\,\mathrm{sinc}^2(2Bt) = m^2(t)$$

Soru: Bir büyük taşıyıcılı çift yan bantlı genlik modülasyonu (large carrier double side band amplitude mod) işareti

$$y(t) = \underbrace{\left[20 + 2\cos(3000\pi t) + 10\cos(6000\pi t)\right]}_{w(t)}\cos(2\pi 10^5 t)$$

ile verilmektedir. Modülasyon indeksi $\mu = 0,6$'dır.

a) $y(t)$'nin spektrumunu bulun ve çizin.

b) Taşıyıcı gücünü, yan bantların gücünü ve verimi bulun.

c) Mesaj işaret $m(t)$'yi bulun.

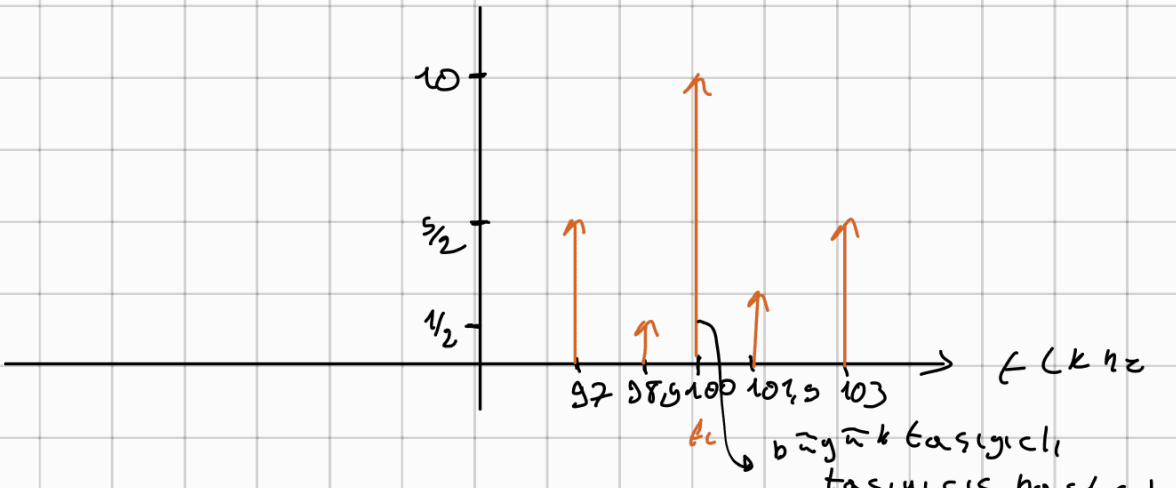
$$y(t) = A_c\left[1 + \mu\, m_n(t)\right]\cos(2\pi f_c t)$$

$$y(t) = w(t)\cos(2\pi 10^5 t)$$

$$\xrightarrow{F} \quad Y(f) = \frac{1}{2}W(f-10^5) + \frac{1}{2}W(f+10^5)$$

$$W(f) = 20\,\delta(f) + \delta(f-1500) + \delta(f+1500) + 5\,\delta(f-3000) + 5\,\delta(f+3000)$$

$$Y(f) = 10\,\delta(f-10^5) + 10\,\delta(f+10^5) + \frac{1}{2}\delta(f-101500) + \frac{1}{2}\delta(f+98500) + \frac{1}{2}\delta(f+101500) + \frac{5}{2}\delta(f-103000) + \frac{5}{2}\delta(f+97000) + \frac{5}{2}\delta(f-97000) + \frac{5}{2}\delta(f+103000)$$



1. yol

$$P_c = \underset{\text{carrier}}{2(10)^2} = 200$$

$$P_s = 4\cdot\left(\frac{5}{2}\right)^2 + 4\left(\frac{1}{2}\right)^2 = 26$$

$$\text{Verim} = \frac{P_s}{P_s+P_c} = \frac{26}{226} = 0{,}115$$

2. yol

$$y(t) = \left[20 + 2\cos(2\pi 1500t) + 10\cos(2\pi 3000t)\right]\cos(2\pi f_c t)$$

taşıyıcı — yan bantlar

$$P_c = \frac{20^2}{2} = 200$$

$$P_s = \frac{1}{2}\left(\frac{2^2}{2} + \frac{10^2}{2}\right) = 26$$

c) $$g(t) = A_c[1 + k_a m(t)]\cos(2\pi f_c t)$$

$$= 20\left[1 + \frac{1}{10}\cos(2\pi 1500t) + \frac{1}{2}\cos(2\pi 3000t)\right]\cos(2\pi 10^5 t)$$

$$\frac{m(t)}{m_p} = \frac{10}{6}\left[\frac{2}{20}\cos(3000\pi t) + \frac{10}{20}\cos(6000\pi t)\right]$$

$$= \frac{1}{12}\left[2\cos(3000\pi t) + 10\cos(6000\pi t)\right]$$

m_p — $m(t)$

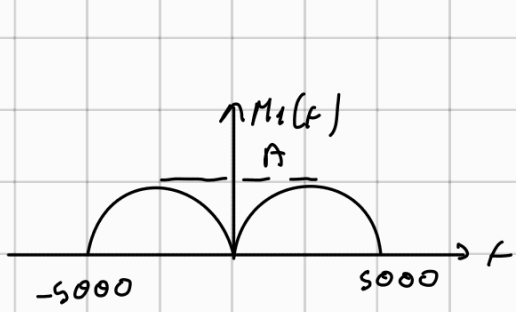
Soru

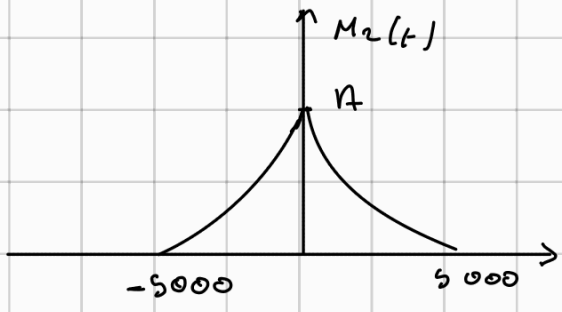
Two signals $m_1(t)$ and $m_2(t)$ both band limited to 5000 Hz, are to be transmitted simultaneously over a channel by multiplexing sheme shown in figure. The signal at point b is the multiplexing signal, which now modulates a carrier freq 10kHz. The modulated signal at point c is transmitted over a channel

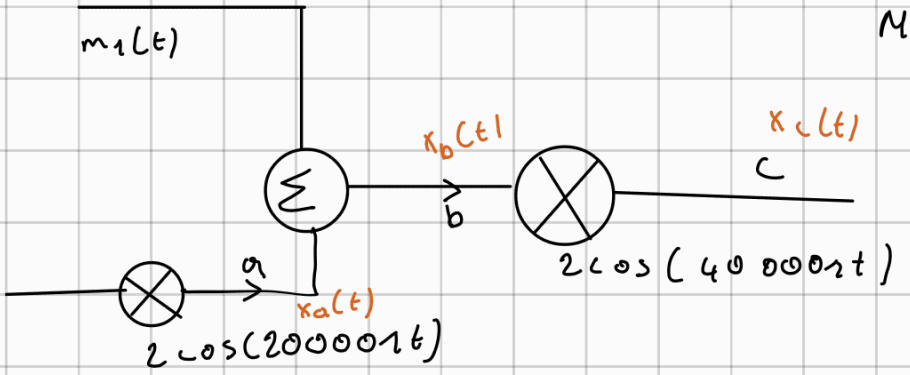
a) Sketch signal spectra (or spectrum) at points a, b, c

b) What must be the bandwith of the channel.

c) Design a receiver to recover signals $m_1(t)$ and $m_2(t)$ from the modulated signal at point c.



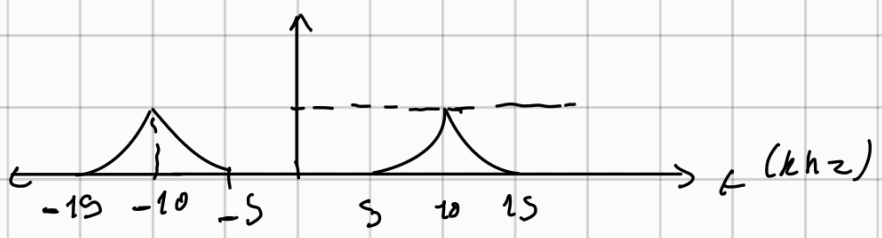




Multiplexing

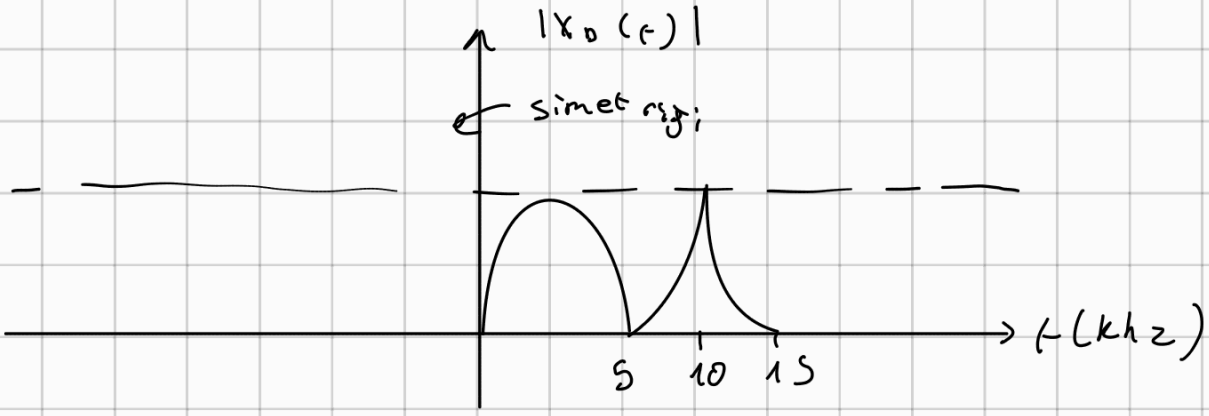
$$x_a(t) = 2m_2(t)\cos(2\pi 10k\,t)$$

$$X_a(f) = M_2(f-10k) + M_2(f+10k)$$



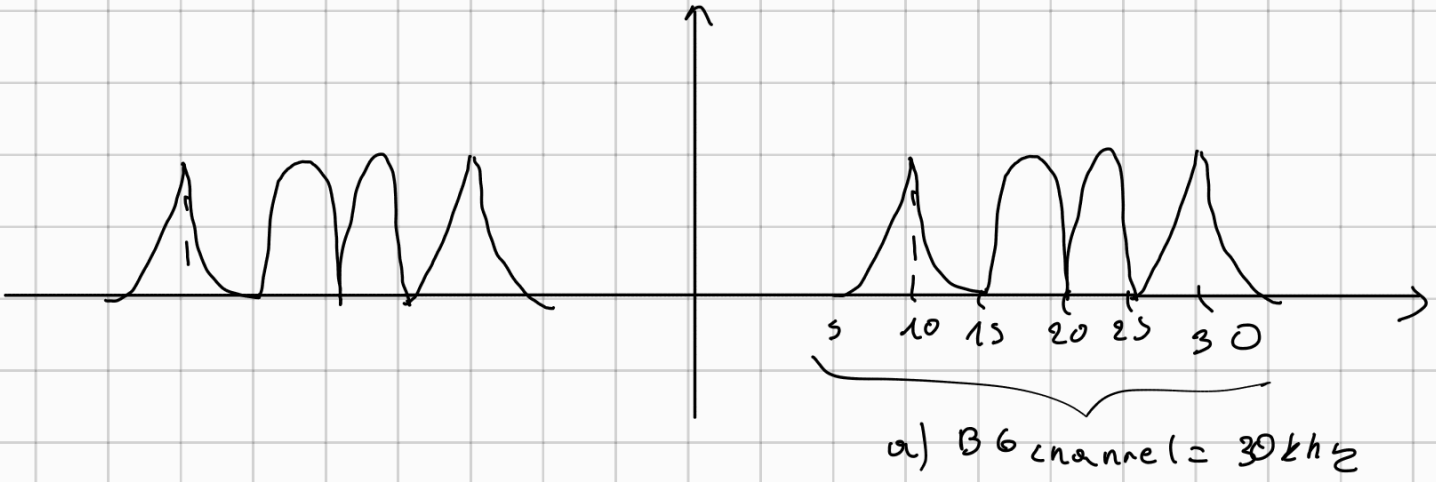
$$x_D(t) = m_1(t) + x_0(t) = m_1(t) + 2m_2(t)\cos(2\pi 10kt)$$

$$X_D(f) = M_1(f) + M_2(f-10k) + M_2(f+10k)$$

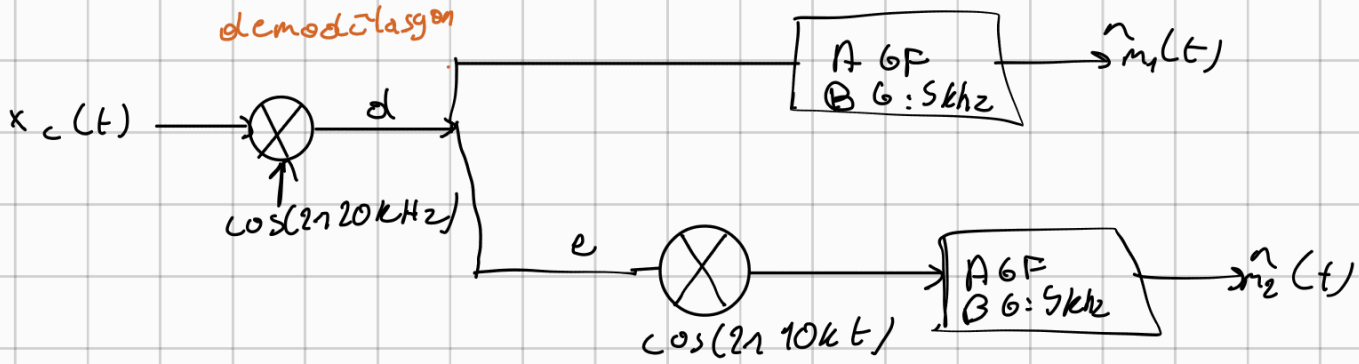


$$x_c(t) = 2x_D(t)\cos(2\pi 20kt)$$

$$X_c(f) = X_D(f-20k) + X_D(f+20k)$$

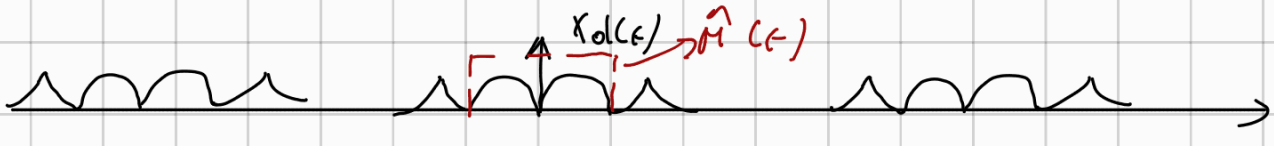


a) BG channel = 30kHz



$$x_d(t) = x_c(t)\cos(2\pi 20kt)$$

$$X_d(f) = \frac{1}{2}X_c(f-20k) + \frac{1}{2}X_c(f+20k)$$



$y_c(t) = x_d(t)\cos(2\pi 10k t) \longrightarrow Y_c(f) = \frac{1}{2} X_d(f-10k) + \frac{1}{2} X_d(f+10k)$

